

Q1. Why are PIC microcontrollers used for most general-purpose applications?

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Through e-mail

A1. PIC (peripheral interface controller) is a product of Microchip Technologies, USA. Salient features of PIC microcontrollers are:

1. Wide range: More than 200 MCUs in the product portfolio. The devices are available for general-purpose as well as specific applications
 2. Available in 8-, 16- and 32-bit in analogue as well as digital signal controller family
 3. RISC-based architecture
 4. Pin-out compatibility
 5. Software compatibility, simple instruction set
 6. Peripheral compatibility
 7. Easy migration across product families
 8. Common and easy-to-use development tools. MPLAB IDE is freely downloadable from website 'www.microchip.com'
 9. Extensive support—information is available in wide range and depth
 10. Available in all package types; for example, PIC16F877A is available in PDIP (with leads) and PLCC for SMD application
- Flexibility of a PIC microcontroller can be well understood from the features of 16F877A. It is an 8-bit microcontroller with 14x8k bits of Flash ROM and 512 bytes of data memory. It has five ports with 33 input/output (I/O) pins, of which eight are analogue and, if need arises, can be converted into digital I/O pins. It has a number of peripherals that give the device immense flexibility:
1. Three timers: 8-bit timer 0, 16-bit timer 1, both of which can be configured as a timer as well as counter. 8-bit timer 2 for PWM operation
 2. Two capture, compare and PWM modules
 3. Master serial synchronous port, which can be configured in SPI or I²C mode for serial communication
 4. Universal synchronous asynchronous receive and transmit (USART)

5. Parallel slave port
6. 10-bit analogue-to-digital converter
7. 256-byte EEPROM
8. Two analogue comparators
9. Capture compare and PWM module

Besides these, features like debugging facility, brown-out reset, watch dog timer and option of choosing the oscillator enhance the functioning of the device.

The user can develop applications by using MPLAB IDE, MPICD3 and PIC development board. Programming can be done either in Assembly language using 35 simple instructions or in 'C' language using a compiler. In fact, it provides the right base to a beginner for learning almost all the facets of a microcontroller. EFY Tech Centers run a 10-day hands-on training course in which all the aspects of this device are covered.

Q2. What is a smartphone? How does it differ from an ordinary phone?

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A2. A smartphone too is a mobile phone but it has more computing abilities and connectivity. You can think of a smartphone as a miniature computer that can also make and receive calls. On account of integrated advance computing, it allows users to install and run more advanced applications based on a specific platform. One of the main features of a smartphone is that you can continue to add applications to the device. Smartphones run complete operating system software providing a platform for application developers.

In a nutshell, a smartphone, besides making telephone calls, has features that are available on a personal digital assistant (PDA) or computer—such as the ability to send and receive e-mail and edit Office documents. It is a handheld device that integrates a cellphone with a PDA, camera, MP3 player, gaming console, or any other information or entertainment device.

While there is no standard definition of the term 'smartphone,' following features set them apart from



Nokia N900 smartphone

normal cellphones:

Operating system. In general, a smartphone has an operating system (OS), while cellphones don't have operating system at all. Smartphones can be powered by following OS: Windows Mobile, iPhone OS, Google's Android, Symbian OS, RIM's BlackBerry, Palm's WebOS and Linux.

Software. While almost all cellphones include some sort of software, a smartphone has the ability to do more. It may allow you to create and edit Microsoft Office documents—or at least view the files. You may download applications, such as personal and business finance managers. Or it may allow you to edit photos and create a playlist of digital tunes.

Web access. More smartphones can access the Web at faster speeds, thanks to the growth of 3G data networks and the addition of Wi-Fi support to many handsets.

Messaging. All cellphones can send and receive text messages, but what sets a smartphone apart is its handling of e-mail. Smartphones can be synced with the e-mail server of your personal or corporate provider. Some smartphones can support multiple e-mail accounts. Others include access to the popular instant messaging services, like AOL's AIM and Yahoo! Messenger.

Network connectivity. Bluetooth, IrDA, GPRS, GPS, Wi-Fi, 3G, FM radio, etc

Hardware available. Accelerometer, gyroscope, MicroSD, sensors, etc.

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